

Vigneshwaran C. J.

AI / ML Engineer and Full-Stack Developer

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PROFILE

Full-stack developer and AI / ML engineer with an M.Tech in Big Data Biology. Work spans desktop agents, enterprise applications, WhatsApp automation, real-time networking, and computational biology tools.

EXPERIENCE

Junior Software Developer | Ceiyone Tech Works Pvt. Ltd.

Dec 2025 - Present

Designing, developing, and maintaining enterprise-grade applications and AI-powered platforms - including AIORA, a multi-tenant WhatsApp business platform of 14 services (bookings, storefronts, n8n-driven AI conversations) running on a shared Hostinger VPS under systemd/PM2 behind nginx, with Render- and Railway-hosted front ends. Work spans backend APIs, WhatsApp automation, cloud deployments, AI/LLM integration, and internal monitoring. Also mentoring 2 engineering interns on the AIORA platform.

AI Engineer Intern | Ceiyone Tech Works Pvt. Ltd.

Oct 2025 - Dec 2025

Built AI-powered automation solutions, OCR-based document systems, LinkedIn automation tools, and multi-agent LLM systems. Integrated AI services, external APIs, and workflow automation platforms.

Research Intern | Manipal Institute of Technology

Jun 2024 - Jul 2024

Developed ML-based QSPR models to predict the solubility of Posaconazole in Self-Emulsifying Drug Delivery Systems (SEDDS). Used RDKit for molecular descriptor generation (50+ descriptors), VIF analysis for feature selection (10 key descriptors: ELUMO, EHOMO, MaxQ-, MaxQ+, MinPartialCharge, MaxPartialCharge, MinAbsPartialCharge, FpDensityMorgan2, PEOE_VSA1, HallKierAlpha). Evaluated Linear Regression, Random Forest, PLS, KNN, and SVM models.

Tutor | Kamakshi Tuition Centre

Jan 2023 - Jun 2023

Taught mathematics to 9th grade students and Python programming to 12th grade students at Kamakshi Tuition Centre, Mugalivakkam Arasa Maram.

SELECTED PROJECTS

PocketPet

One cross-platform product comprising a Tauri 2/Rust desktop companion and its React 19/TypeScript download website. The desktop app provides JavaScript/SVG pets, games, customisation, break reminders, and a streamed task agent that searches, reads pages, and controls a dedicated Chromium profile via CDP. It supports Anthropic and OpenAI-compatible providers including Ollama, with approval gates, secret-field refusal, spend limits, scheduling, and cancellation. The website reuses the app's sprites as an interactive demo, explains platform support, reads GitHub Releases, and directs visitors to the correct Windows...

Technologies: Rust, Tauri, JavaScript, React.js, TypeScript, Vite, Tailwind CSS, Tokio, Win32, Chrome DevTools Protocol, Agentic AI, GitHub API, GitHub Actions, Vercel

Syncly

Developer-to-developer networking platform built solo end to end. Members pick a mentorship role (mentor, learner, both), swipe a paginated feed of developers they have not met (Vibe to connect, Ghost to pass); the recipient replies Link or Noped, and a mutual Link unlocks real-time 1:1 chat over Socket.IO with persisted history. Each member gets a shareable portfolio page at their slug (projects, experience, education, certifications, theme, SEO, visibility). httpOnly JWT cookie auth with bcrypt; AWS SES email on every swipe plus a daily 08:00 node-cron reminder for pending Vibes. Loop: Discover to Vibe to Link to ...

Technologies: React 19, Vite, Redux Toolkit, React Router 7, Tailwind CSS 4, daisyUI, Axios, Node.js, Express 5, MongoDB, Mongoose, Socket.IO, JWT, bcrypt, AWS SES, node-cron, AWS EC2

GlycanBench: a unified resource for working with glycans

Full-stack glycoinformatics platform and installable Python package (glycanbench: SDK, CLI, server) built under Ashok Palaniappan at SASTRA's Systems Computational Biology Lab; live at glycanbench.sastra.edu. Eight tool groups behind one React UI and one FastAPI API: Create (click-to-build glycans with SNFG 2D + 3D conformers via ETKDGV3/MMFF94s, Cytoscape.js biosynthetic networks, IUPAC/WURCS/GlycoCT/SMILES conversion), Visualize (2D SNFG, 3Dmol.js, KEGG pathways), Analyse (characterisation, GlyTouCan insight, RDKit descriptors + fingerprints, motif mutation), Compare (Tanimoto over 5 fingerprints), Align (Needl...

Technologies: Python, FastAPI, Pydantic, React 19, TypeScript, Vite, Tailwind CSS 4, PyTorch, PyTorch Geometric, RDKit, glycowork, glypy, BioPython, scikit-learn, SciPy, LangChain, Groq, 3Dmol.js, Cytoscape.js, Chart.js

EDUCATION

Master of Technology in Big Data Biology

SASTRA Deemed University | Aug 2023 - Jun 2025 | 79.67%

Bachelor of Technology

Prathyusha Engineering College | Aug 2019 - Apr 2023 | 84.4%

TECHNICAL SKILLS

Programming Languages: Python, R, SQL, JavaScript, TypeScript, Rust

Data Science & ML: Pandas, NumPy, Scikit-Learn, PyTorch, TensorFlow, SHAP

Frontend Development: React.js, Next.js, Redux, HTML5, CSS3, Tailwind CSS, Bootstrap

Backend Development: Node.js, Express.js, FastAPI, Flask, Streamlit, Socket.IO, BullMQ, Mongoose, Tauri

Databases: MongoDB, PostgreSQL, DynamoDB, Oracle, Redis

Cloud & Devops: AWS EC2, AWS SES, Google Cloud, Firebase, Docker, PM2, systemd, nginx, Render, Railway, GitHub Actions

Ai & Llms: OpenAI GPT-4o, LangChain, LangGraph, Groq, Agentic AI

Bioinformatics & Research: RDKit, Mordred, glycowork, PyTorch Geometric, BioPython, Glycomics, Cytoscape, Galaxy

Tools: Git, GitHub, n8n, Postman, Playwright, Jupyter Notebook, Linux, Vim, Arduino, Photoshop, macOS

CERTIFICATIONS

Namaste ReactJS | NamasteDev

Namaste Node.js | NamasteDev